

HERNANDO, PANTINE B.

BSIT-3

CHALLENGE – 13 - CRYPTOGRAPHY

rail-fence 

Medium Cryptography picoCTF 2022

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Description

A type of transposition cipher is the rail fence cipher, which is described [here](#). Here is one such cipher encrypted using the rail fence with 4 rails. Can you decrypt it?

Download the message [here](#).

Put the decoded message in the picoCTF flag format,

`picoCTF{decoded_message}`.

Hints 

1

19,023 users solved



86%

Liked



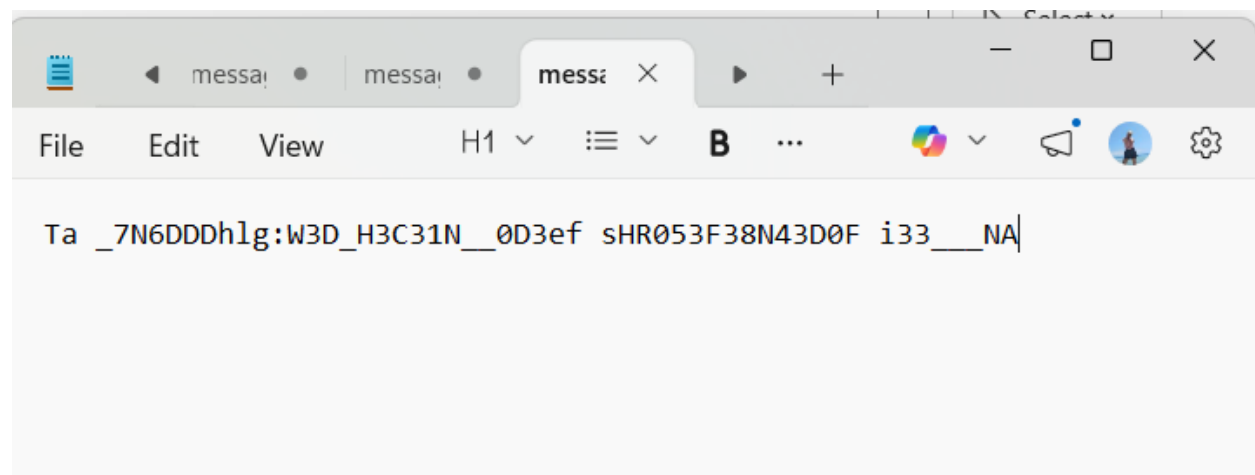
picoCTF{FLAG}

Submit
Flag

picoCTF{decoded_message} flag format.

2. Identify the Ciphertext

The encrypted message provided in the challenge file is:



Solution

Walkthrough

1. Analyze the Requirements

The challenge description specifies that the message was encrypted using a **rail fence cipher** with **4 rails**. The goal is to decrypt the provided ciphertext and format it into the standard

3. Decryption Process

To solve this, a decryption tool (Boxentriq) was utilized. The following parameters were applied to the tool:

- **Mode:** Decrypt
- **Rails:** 4
- **Offset:** 0

The screenshot shows the Boxentriq website's Rail Fence Cipher tool. The browser address bar shows 'boxentriq.com/ciphers/rail-fence-cipher'. The page has a sidebar on the left with links: 'On this page', 'Start', 'Rail Fence Cipher', 'Features', 'Options', 'What is the Rail Fence cipher?', 'How it works', 'History and facts', and 'Where You'll Encounter It'. The main content area is titled 'Manual mode: Enter text, choose rails and offset, then encode or decode.' It contains three sections: 'Input (ciphertext)' with a text area containing 'Ta _7N6DDdhlg:W3D_H3C31N__0D3ef SHR053F38N43D0F i33__NA' and an 'Options' button; 'Mode', 'Rails', and 'Offset' settings, all set to 'Decrypt', '4', and '0' respectively, with explanatory text below each; and 'Output (plaintext)' with a text area showing 'The flag is: WH3R3_D035_7H3_F3NC3_8361N_AND_3ND_D00AFDD3' and an 'Options' button. At the bottom, a 'Rail layout' section shows a zig-zag diagram of the ciphertext characters across four rails.

BOXENTRIQ

Manual mode: Enter text, choose rails and offset, then encode or decode.

Input (ciphertext) Options

Ta _7N6DDdhlg:W3D_H3C31N__0D3ef SHR053F38N43D0F i33__NA

Mode Decrypt ▾ **Rails** 4 **Offset** 0

Encrypt = Plaintext to ciphertext or Decrypt = ciphertext to plaintext. Number of zig-zag rails to write on. Starting rail (0 = top).

Output (plaintext) Options

The flag is: WH3R3_D035_7H3_F3NC3_8361N_AND_3ND_D00AFDD3

Rail layout ^

Shows how the text moves in a zig-zag across the rails.

```
T   a       _   7   N   6   D   D   D
h   l   g   :   W   3   D   _   H   3   C   3   1   N   _   _   0   D   3
e   f       s   H   R   0   5   3   F   3   8   N   4   3   D   0   F
           i   3   3   _   _   _   N   A
```

By placing the ciphertext into the tool, the "zig-zag" pattern of the rail fence was reconstructed to reveal the original plaintext.

4. Resulting Plaintext

The decryption tool produced the following output:

The flag is: **WH3R3_D035_7H3_F3NC3_8361N_AND_3ND_D00AFDD3**

5. Final Flag

Following the required picoCTF format:
picoCTF{WH3R3_D035_7H3_F3NC3_8361N_AND_3ND_D00AFDD3}

Reflection

The Rail Fence cipher is a classic example of a **transposition cipher**, where the characters of the message remain the same, but their order is rearranged based on a specific geometric path. This challenge reinforces the importance of identifying the correct "key"—in this case, the number of rails—to successfully reassemble the message.

Using automated tools like Boxentriq is highly efficient for CTFs, but the underlying logic remains the same: mapping characters back from their "rail" positions into a linear string. This task highlights how even simple historical ciphers can become complex to read manually once the number of rails increases, yet they remain vulnerable to basic cryptanalysis once the rail count is known or brute-forced.